

Agri-Sovereign-2B: A Morphologically Extended Continual Pre-Training Foundation Model for Hyper-Local Agro-Ecological Reasoning

Foundational Context and Regional Imperative

Deploying general-purpose artificial intelligence within specialized regional sectors has historically been impeded by structural misalignments between global foundation models and vernacular sociotechnical domains. In the agricultural heartland of Tamil Nadu, particularly the agrarian corridor of the Coimbatore and Pollachi basins surrounding Kondampatty, smallholder farming enterprises face compound vulnerabilities. These challenges include climate-induced shifts in the southwest and northeast monsoons, ground water depletion, and severe localized outbreaks of virulent phytopathologies such as the fall armyworm (*Spodoptera frugiperda*) in maize and the rugose spiraling whitefly (*Aleurodicus rugioperculatus*) in coconut groves. While state-of-the-art foundation models demonstrate proficiency in standardized Anglo-centric tasks, their raw deployment in vernacular agronomy fails systematically. This failure is governed by two computational and epistemic barriers: the token fertility penalty inherent to Byte-Pair Encoding (BPE) algorithms applied to Dravidian scripts, and pervasive agronomic hallucination that contradicts established agro-climatic protocols.

Agri-Sovereign-2B is conceived as an indigenous, domain-adapted bilingual foundation model engineered to resolve these structural pathologies. By expanding the underlying lexical embedding space with linguistically informed Tamil morphemes and executing continual pre-training across high-density agronomic corpora on institutional NVIDIA tensor-core clusters, this initiative bridges the gap between high-performance computing infrastructure and vernacular agricultural impact.

1. Problem Statement

The inability of existing large language models to function as dependable agronomic co-pilots in rural Tamil Nadu is rooted in two architectural and systemic deficiencies:

The Indic Token Fertility Penalty and Computational Inefficiency

Modern large language models utilize subword tokenizers optimized almost exclusively for the statistical distributions of Latin-script European languages. Tamil is an agglutinative, morphologically rich language where grammatical relations, tense, case markers, and modality are concatenated directly onto lexical roots. Standard multilingual tokenizers, such as those embedded within Meta's Llama series or OpenAI's GPT architectures, possess virtually no multi-character token representations for Tamil phonemes and bound morphemes. Consequently, these tokenizers disintegrate complex Tamil words into fine-grained character fragments or raw byte-level representations. This fragmentation leads directly to an extreme

elevation in token fertility, defined as the ratio of generated tokens to natural words ($\tau = \frac{N_{\text{tokens}}}{N_{\text{words}}}$). Whereas English text yields an average fertility rate between 1.2 and 1.4 tokens per word, contemporary foundation models exhibit fertility rates between 4.4 and 11.4 for Tamil.

Foundation Model / Tokenizer Architecture	Mean Tamil Fertility (τ_{ta})	Mean English Fertility (τ_{en})	Effective Context Window Degradation Factor	Relative KV Cache Memory Inflation
OpenAI GPT-4 / GPT-4o	11.41	1.28	8.91\times	8.91\times
Meta Llama-3-8B	11.40	1.32	8.64\times	8.64\times
Meta Llama-2-7B	10.66	1.36	7.84\times	7.84\times
Google Gemma-7B	4.39	1.34	3.28\times	3.28\times
Sarvam-1 (Indic Custom BPE)	1.82	1.35	1.35\times	1.00\times
Agri-Sovereign-2 B (Proposed Morpheme-BPE)	1.74	1.31	1.29\times	0.96\times

This systemic fertility penalty exerts severe downstream performance penalties during training and inference. An 8,192-token context window accommodates roughly 6,000 words in English, but collapses to fewer than 750 words in Tamil when processed through standard Llama-style tokenizers. In multi-turn retrieval or diagnostic settings, context saturation occurs almost immediately.

Furthermore, autoregressive decoding latency scales linearly with token sequence length, forcing the hardware to execute up to eight times more forward matrix-vector passes to generate an equivalent paragraph of advice. This behavior consumes excessive memory bandwidth and multiplies Key-Value (KV) cache allocation. At the representational level, premature byte fragmentation forces the initial layers of the neural network to expend parameter capacity reassembling basic phonetic syntax rather than modeling agronomic logic.

Epistemic Misalignment and Agronomic Hallucination

Contemporary generalist models lack deep parametric knowledge regarding tropical agro-ecological systems, statutory agricultural pesticide compliance, and regional soil epidemiology. When evaluated on vernacular queries regarding local crops, generalist models frequently fabricate dosage calculations, suggest agrochemicals banned by the Central Insecticides Board & Registration Committee (CIBRC), or propose treatments unsuited to the red loam and black cotton soils characterizing the Coimbatore and Pollachi belts. The absence of localized grounding renders these models unsafe for real-world agronomic deployment without rigorous parametric domain adaptation.

2. Technical Objectives

The project is governed by five tightly coupled technical milestones:

- **Morphemic Tokenizer Reconstruction:** Induce an agglutinative vocabulary expansion comprising 24,000 domain-specific and structural Tamil morpheme tokens, compressing baseline Tamil fertility from $\tau \approx 11.4$ down to $\tau \leq 1.85$ tokens per word while preserving original English tokenizer representations.
- **Low-Divergence Latent Embedding Surgery:** Align the expanded embedding layer using Hewitt semantic-anchor projection, initializing new tokens as linear combinations of constituent historical embeddings to prevent representational collapse during early training.
- **Distributed Continual Domain Pre-Training (CPT):** Execute a two-stage continual pre-training curriculum across 4.5 billion tokens using NVIDIA NeMo and PyTorch Fully Sharded Data Parallel (FSDP-2), infusing native Tamil syntactic structures and institutional TNAU agronomic corpora into the model weights.
- **Five-Part Actionable Agronomic Alignment:** Supervised fine-tuning across curated, expert-validated instruction datasets enforcing an actionable structure consisting of Symptom Validation, Immediate Intervention, Scientific Rationale, Preventative Practice, and Extension Referral.
- **Optimized High-Throughput Edge Deployment:** Quantize and package the resulting foundation model into INT4/FP8 formats via TensorRT-LLM and vLLM, securing inference throughput exceeding 60 tokens per second per stream on accessible institutional and edge hardware.

3. Existing Approaches and Baseline Limitations

Efforts to adapt artificial intelligence to Indian regional contexts have historically followed three primary approaches, all of which exhibit fundamental architectural trade-offs:

Retrieval-Augmented Generation Over Frozen Generalist LLMs

The most common enterprise pattern connects a frozen, closed-source API (such as OpenAI GPT-4o) or a large open-weight checkpoint (such as Llama-3.1-8B) to an external vector database populated with translated agricultural texts. While this eliminates pre-training compute requirements, it amplifies inference-time token fertility costs.

Retrieving multilingual dense vectors often introduces cross-lingual retrieval failure when colloquial Tamil farming dialects are matched against formal academic publications.

Furthermore, stuffing retrieved passages into a frozen prompt inflates context token consumption by an order of magnitude, resulting in high API operational expenses and serving latencies unsuited for interactive rural deployment.

Parameter-Efficient Fine-Tuning Without Tokenizer Restructuring

Standard academic initiatives frequently execute Low-Rank Adaptation (LoRA) or QLoRA on generalist foundation models using small sets of translated instruction pairs. This approach leaves the underlying tokenization structure untouched, meaning the model continues to inherit high sequence length overhead and KV-cache expansion.

Furthermore, low-rank parameter updates alter only a fraction of the weight matrices, which is insufficient to inject deep, factual semantic graphs covering extensive agrochemical nomenclatures, botanical classifications, and regional pest life-cycles. The resulting models frequently master the stylistic tone of conversational Tamil while continuing to hallucinate core technical facts.

Monolithic Multilingual Pre-Training from Scratch

Pioneering foundation projects such as Sarvam-1 demonstrate that a dedicated 2-billion parameter model trained from scratch can surpass significantly larger architectures across Indic tasks. Sarvam-1 achieved state-of-the-art results across Indic benchmarks while securing fertility rates of 1.4 to 2.1 by pre-training on 4\text{ trillion} tokens. However, full pre-training requires immense supercomputing scale, utilizing 1,024 NVIDIA H100 GPUs across continuous five-day allocations. For academic institutions operating individual dedicated NVIDIA multi-GPU racks, training a foundational model from scratch on multi-trillion token datasets is computationally intractable within hackathon timelines.

The Continual Pre-Training Paradigm

Agri-Sovereign-2B resolves these contradictions through Continual Pre-Training with targeted vocabulary expansion. By inheriting the reasoning, logic, and base representations of a compact foundation checkpoint, expanding its lexical matrix, and continually updating all layers on a concentrated, high-quality domain corpus, the model achieves the low token fertility and deep linguistic fluency of a bespoke foundation model at a fraction of the compute budget.

Paradigm	Architectural Approach	Token Efficiency (Tamil)	Parametric Domain Grounding	Compute Requirement	Hardware Target Alignment
RAG on Frozen Base	Static Base + Vector Search	Unfavorable ($\tau > 10$)	Ephemeral / Contextual	Low / Zero Training	Cloud API Consumer
LoRA Fine-Tuning	Parameter-Efficient Adapter	Unfavorable ($\tau > 10$)	Superficial Styling	Minimal (1 GPU-day)	Desktop / Single Card
From-Scratch Pre-Training	Random Weight Initialization	High ($\tau < 2.0$)	Broadly Distributed	Extreme (1024 GPUs, 5\text{d})	Hyperscale Datacenter
Agri-Sovereign-2B (CPT)	Vocab-Expanded Full CPT	High ($\tau = 1.74$)	Deep Agro-Ecological	Balanced (8 GPUs, 15\text{h})	Institutional NVIDIA Rack

4. Input Data Architecture: Composition, Volume, and Curation

The data strategy utilizes a multi-tier curriculum designed to progressively establish morphological fluency, general cross-domain reasoning, and deep agronomic diagnostics without triggering catastrophic forgetting of core logical primitives. The continual pre-training corpus is structured in two successive training regimes, followed by a dedicated post-training alignment phase. The initial phase focuses on establishing linguistic and syntactic grounding across modern Tamil, utilizing large verified corpora to adapt the newly

expanded embedding weights. The subsequent phase introduces dense technical, biochemical, and agricultural texts to shift the internal representations toward actionable agronomy. Finally, the model undergoes supervised fine-tuning and preference optimization using structured agricultural instruction pairs.

Curriculum Phase	Dataset Name & Primary Origin	Uncompressed Size	Curated Token Count	Domain and Linguistic Representation
Phase 1: Syntactic CPT	AI4Bharat IndicCorp v2 (Tamil)	14.2\text{ GB}	1.80\text{ Billion}	Formal Tamil syntax, news, governmental publications
Phase 1: Syntactic CPT	AI4Bharat Sangraha Verified (Tamil)	6.5\text{ GB}	0.80\text{ Billion}	Verified high-quality web text, humanities, sciences
Phase 1: Syntactic CPT	FineWeb-Edu & OpenWebMath Replay	1.6\text{ GB}	0.40\text{ Billion}	General math, functional logic, and algorithmic replay
Phase 2: Domain CPT	TNAU Agritech Portal Documentation	4.1\text{ GB}	0.50\text{ Billion}	Crop protection guides, pest lifecycle databases, IPM
Phase 2: Domain CPT	ICAR-CRIDA & IMD Agro-Bulletins	2.2\text{ GB}	0.30\text{ Billion}	Weather risk contingencies, soil health, drip irrigation
Phase 2: Domain CPT	Synthetic Agronomic Diagnostic Chains	3.2\text{ GB}	0.40\text{ Billion}	Multi-step phytopathology reasoning and diagnostic traces
Phase 2: Domain CPT	AI4Bharat Samanantar (Agrl. Parallel)	2.8\text{ GB}	0.30\text{ Billion}	Parallel technical pairs for concept space alignment
Total Continual Pre-Training	Aggregated Pre-Training Pipeline	34.6\text{ GB}	4.50\text{ Billion}	Complete CPT Representation Space
Phase 3: SFT & DPO	TNAU Extension & KCC Structured Sets	0.9\text{ GB}	0.12\text{ Billion}	Supervised instruction dialogues and preference pairs

Data Curation, Filtering, and Decontamination

Raw web extractions and scanned extension documentation undergo rigorous automated filtering prior to tokenization. Near-duplicate documents are eliminated using MinHash Locality-Sensitive Hashing with a Jaccard similarity threshold of 0.80, followed by document-level language identification to exclude non-standard romanized dialect fragments.

Text quality is further validated using a fast n-gram KenLM language model trained on classical and contemporary literature, pruning text segments that yield excessively high perplexity scores indicative of garbled OCR output or template noise.

To prevent benchmark contamination, an automated n-gram decontamination pipeline evaluates the entire pre-training corpus against the evaluation sets of IndicMMLU, IndicGenBench, and the custom Agro-Tamil Diagnostic Suite. Any document exhibiting an exact 13-gram overlap with the validation corpora is purged from the pre-training pool.

Instruction fine-tuning pairs sourced from Kisan Call Centre logs and agricultural extension leaflets are structured into a standardized five-part advisory framework. This structure mandates that every response begin with an explicit acknowledgment of the crop symptom and district agro-ecological zone, detail the exact chemical or biological intervention, state the physiological rationale, provide long-term soil management practices, and conclude with the contact details for the local Krishi Vigyan Kendra or TNAU research station.

5. Proposed Model Architecture and Engineering Pipeline

Agri-Sovereign-2B is engineered as an autoregressive decoder-only transformer initialized from high-performing open-weight checkpoints such as **Qwen-2.5-1.5B** or **Llama-3.2-3B**. Research indicates that deeper, narrower parameter geometries deliver superior transfer capabilities and parameter efficiency over shallow, wide configurations during cross-lingual domain adaptation.

Architectural Feature	Technical Specification	Operational Rationale
Base Configuration	Dense Decoder-Only Transformer	Standardized autoregressive causal language modeling
Transformer Layers (L)	28 Hidden Layers	Deep-and-thin design optimizing parameter representations
Hidden Dimension (d_{model})	2,048	Balanced representational capacity and memory footprint
Feed-Forward Dimension (d_{ffn})	11,008 (SwiGLU Activation)	High non-linear expressive capacity across token projections
Attention Mechanism	Grouped-Query Attention (GQA)	16 Query heads, 8 Key-Value heads; reduces KV-cache latency
Position Embeddings	Rotary Position Embeddings (RoPE)	Scaled base frequency ($\theta = 500,000$) for long-context stability
Context Window (L_{ctx})	8,192 Tokens	Full support for complex retrieved agro-climatic histories
Base Vocabulary Size	128,000 Tokens	Native multilingual vocabulary (English, code, multilingual)
Added Tamil Morphemes	24,000 Dedicated Tokens	Linguistically induced subwords, roots, and case markers
Final Vocabulary Size	152,000 Tokens	Unified representation

Architectural Feature	Technical Specification	Operational Rationale
		minimizing Tamil token fertility
Normalization Protocol	RMSNorm ($\epsilon = 10^{-6}$)	Pre-normalization ensuring gradient stability across CPT

Agglutinative Vocabulary Induction and Embedding Alignment

Expanding the vocabulary from 128,000 to 152,000 tokens requires surgical initialization to prevent catastrophic disruption of the pre-existing embedding space. Rather than assigning random Gaussian weights to the 24,000 newly allocated token slots, the model utilizes Hewitt semantic-anchor initialization.

For each new Tamil morpheme v_{new} , the string is tokenized by the base pre-trained tokenizer into its constituent subwords or bytes $S(v_{\text{new}}) = \{s_1, s_2, \dots, s_k\}$. The initial embedding vector $E_{\text{init}}(v_{\text{new}})$ is computed as the arithmetic mean of the existing embeddings of its constituent components:

$$E_{\text{init}}(v_{\text{new}}) = \frac{1}{|S(v_{\text{new}})|} \sum_{s \in S(v_{\text{new}})} E_{\text{orig}}(s)$$

This projection guarantees that new tokens begin training positioned close to their semantic equivalents within the pre-trained latent space, minimizing initial Kullback-Leibler divergence. During the initial 1,500 optimization steps of Phase 1, the transformer backbone weights are held static, calculating backpropagation updates solely for the input embedding matrix E and the output language modeling head W_{lm} . This initial warm-up anchors the expanded vocabulary into the latent distribution before unfreezing all 28 transformer layers for full continual pre-training.

6. Model Parameter Scale and Compute Budget Justification

The choice of a 2.1B-parameter configuration balances deep reasoning capacity with rigorous compute feasibility on dedicated institutional hardware. Sizing the model appropriately ensures full execution within the operational boundaries of a hackathon timeline while maximizing hardware throughput.

Mathematical FLOP Scaling Formulation

Following standard transformer scaling laws, the floating-point operations (FLOPs) required to train a causal decoder model containing N parameters across D tokens are approximated by:

$$C \approx 6 \times N \times D \quad \text{FLOPs}$$

For Agri-Sovereign-2B ($N \approx 2.1 \times 10^9$ non-embedding parameters) continually pre-trained across $D = 4.5 \times 10^9$ tokens (4.5Billion tokens) total across Phases 1 and 2):

$$C = 6 \times (2.1 \times 10^9) \times (4.5 \times 10^9) = 5.67 \times 10^{19} \quad \text{FLOPs}$$

Cluster Hardware Throughput and Runtime Analysis

Institutional rack installations typically incorporate nodes equipped with eight enterprise GPUs interconnected via high-bandwidth NVLink. The operational feasibility of completing Phase 1

and Phase 2 continual pre-training within realistic timeframes is detailed below, assuming realistic Model FLOPs Utilization (MFU) using FlashAttention-3 / FlashAttention-2 and mixed-precision execution.

Institutional Cluster Configuration	Theoretical Peak BF16 Compute per GPU	Achievable Sustained MFU	Total Sustained Compute Rate	Wall-Clock Time for 4.5B Token CPT
Node: 8\times NVIDIA A100-SXM4-80GB	312\text{ TFLOPs}	42\%	1.048\time[span_182](start_span)[span_182](end_span)s 10^{15}\text{ FLOPs/s}	15.01\text{ Hours}
Node: 8\times NVIDIA H100-SXM5-80GB	989\text{ TFLOPs}	44\%	3.481\time[span_183](start_span)[span_183](end_span)s 10^{15}\text{ FLOPs/s}	4.52\text{ Hours}
Node: 8\times NVIDIA L40S-48GB	366\text{ TFLOPs}	38\%	1.112\times 10^{15}\text{ FLOPs/s}	14.16\text{ Hours}

The mathematical scaling demonstrates that continual pre-training across 4.5\text{ billion} tokens requires between 4.5 and 15.0 wall-clock hours on an 8-GPU enterprise node. This leaves substantial compute margin within a multi-day hackathon window for preliminary embedding alignment, hyperparameter sweeps, validation runs, and subsequent instruction fine-tuning.

7. Expected Outcomes and Empirical Evaluation Benchmarks

The project will establish measurable improvements across three evaluation dimensions: subword token efficiency, academic Indic linguistic benchmarks, and verified agricultural decision accuracy.

Quantifiable Performance Targets

The expanded morpheme tokenizer is designed to constrain Tamil fertility to \le 1.85 tokens per word, representing an approximate 6\times compression compared to the baseline Llama-3 tokenizer (11.40). This compression directly reduces generation latency and memory bandwidth consumption during inference, unlocking generation speeds in excess of 60 tokens per second per concurrent user under quantized runtime environments.

Standardized Evaluation Benchmark	Baseline Llama-3.2-3B	Baseline Gemma-2-2B	Sarvam-1 (2B Scratch)	Agri-Sovereign-2B (Target)
IndicMMLU (Tamil Subset)	32.14\%	30.[span_50](start_span)[span_50](end_span)82\%	43.79\%	46.50\%
TriviaQA Indic	46.00\%	40.58\%	90.62\%	91.20\%

Standardized Evaluation Benchmark	Baseline Llama-3.2-3B	Baseline Gemma-2-2B	Sarvam-1 (2B Scratch)	Agri-Sovereign-2B (Target)
(Tamil)				
IndicGenBench (CrossSum)	19.82	6.57	20.48	22.80
Agro-Tamil Diagnostic Suite (Custom)	24.50\%	19.20\%	38.40\%	88.50\%
CIBRC Statutory Dosage Compliance	31.00\%	28.00\%	44.00\%	96.00\%

Custom Validation Suite: Agro-Tamil Diagnostic Suite

To rigorously evaluate real-world utility, an evaluation dataset comprising 500 expert-validated agronomic challenges grounded in TNAU Crop Production Guides will assess performance across three core areas:

- *Phytopathological Identification*: Evaluating the model's capacity to diagnose leaf spots, stem borers, and wilt syndromes described in colloquial Tamil dialect phrasing.
- *Statutory Chemical Compliance*: Measuring the accuracy of recommended chemical active ingredients, ensuring zero recommendations of banned insecticides and strictly verifying that recommended dilution rates per hectare match official CIBRC schedules.
- *Hyper-Local Agro-Ecological Accuracy*: Assessing the system's capacity to contextualize advice based on local geography, such as factoring in the soil types and micro-climatic pest vectors common to the Coimbatore and Pollachi regions.

8. Proposed Product and Application Architecture

The operational manifestation of the model is **Uzhavan-Sahayak**, an end-to-end agronomic advisory system tailored to the rural connectivity constraints of Tamil Nadu.

The user engagement pipeline is built around an omni-channel interface supporting both a lightweight, offline-first Progressive Web Application (PWA) and an automated WhatsApp Business messaging gateway. Farmers submit queries through spoken Tamil voice recordings, text prompts, or field photographs of symptomatic plant foliage. Spoken voice notes are routed into a streaming acoustic front-end powered by an IndicWav2Vec or fine-tuned Whisper-Tamil model, which streams high-fidelity vernacular transcriptions to the central reasoning engine. The core advisory layer is powered by Agri-Sovereign-2B, hosted within an institutional inference environment utilizing vLLM or NVIDIA TensorRT-LLM. The serving engine incorporates PagedAttention and dynamic request batching to maintain low serving latencies across simultaneous requests.

The inference engine dynamically enriches user queries by retrieving real-time localized weather telemetry from the India Meteorological Department (IMD) and weekly Agro-Met bulletins issued by TNAU. Before an autoregressive response is returned to the farmer, it passes through an automated CIBRC safety validation filter that screens the output against structured chemical and dosage tables to verify that recommended compounds and application rates are safe and legally approved.

Once verified, the structured advice is formatted according to the five-part advisory framework

and routed back through the communication gateway. The system simultaneously sends a structured text card and an audio message generated by a regional IndicTTS voice synthesis engine, allowing farmers of varying literacy levels to readily act upon the recommendations.

High-Performance Computing Architecture and GPU Utilization Blueprint

Executing a continuous 4.5-billion token pre-training run requires an enterprise distributed computing architecture configured to maximize hardware utilization:

Distributed Sharding and Parallelism Strategy

The training pipeline uses PyTorch FSDP-2 integrated with the NVIDIA NeMo framework, implementing ZeRO-3 style parameter sharding across the cluster. In this configuration, model parameters, gradient accumulations, and 32-bit AdamW optimizer states are sharded evenly across the eight GPUs of the NVIDIA rack.

Individual layer weights are dynamically reconstructed via all-gather collective operations during the forward pass and released immediately afterward, reducing per-card memory consumption and accommodating a large global batch size of 524,288 tokens through sequence packing without out-of-memory errors.

Kernel Fusion and Memory Bandwidth Optimization

To ensure maximum hardware saturation, all self-attention operations use FlashAttention-3 on Hopper architectures or FlashAttention-2 on Ampere setups. FlashAttention circumvents the quadratic memory scaling of classical attention mechanisms by tiling matrix multiplications and computing softmax reduction asynchronously within GPU SRAM.

This implementation eliminates high-bandwidth memory round-trips for attention matrices, sustaining Model FLOPs Utilization above 40% across the entire multi-billion-token training run.

Precision Scaling and Pipeline Caching

The training runtime runs in native BF16 mixed precision, leveraging FP8 Tensor Core execution for matrix multiplications (GEMMs) where supported to maximize arithmetic throughput without gradient loss.

To prevent data starvation across high-throughput GPU interconnects, raw text data is tokenized offline and organized into binary memory-mapped index structures (.bin and .idx) compatible with Megatron-LM data loaders. This memory-mapped data ingestion pattern bypasses Python runtime garbage collection and CPU deserialization bottlenecks, ensuring that GPU compute units remain fully saturated throughout training.

Proposal Evaluation Summary

The Agri-Sovereign-2B proposal aligns closely with the technical and practical objectives of LLM Forge 2026:

- **Authentic Foundation Engineering:** Rather than presenting an application-level wrapper

around proprietary APIs, this proposal tackles foundational model development—implementing morpheme-aware tokenization, embedding layer alignment, distributed continual pre-training, and scaling optimization.

- **Societal and Institutional Relevance:** Rooted in TNAU research and designed for the agro-ecological conditions of the Coimbatore and Pollachi agricultural belt, the project delivers practical regional value that supports broader sovereign artificial intelligence initiatives.
- **Methodical Compute Feasibility:** Sizing the architecture at 2.1\text{B} parameters and the pre-training corpus at 4.5\text{B} tokens matches the empirical compute capacity of an 8-GPU enterprise node, ensuring that the project can train stably, evaluate rigorously, and deploy successfully within the hackathon operational window.

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